EXPLORING EVALUATIVE LANGUAGE AND SENTIMENT IN GAME REVIEWS: A COMPARATIVE CORPUS STUDY OF SEQUEL GAMES

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INTRODUCTION

In the highly competitive digital game market, game developers use players' opinions to update and improve a game. Besides, potential players may use other gamers' judgements to commit to the purchase. For these reasons, game reviews are crucial for developers and gamers (Miao et al., 2022).

Steam is the largest digital game platform used to purchase games and connect with other gamers and holds a respectable US market share of 75% in 2025 (Kumar, 2026). An important reason for Steams popularity is its transparency: reviews are supported by the amount of playing hours at the time of review, verifying non-sponsored ownership of the game.

Analyzing individual game reviews, however, can be a time-consuming job, especially for popular games: Counter-Strike 2, a top-selling game on Steam, has over 9.5 million reviews, of which 2.5 million in English. It is for this reason that review analyses are automated, as done in the present study.

Automation of game review analysis has developed quickly, using different methods, attempting to build the most accurate analysis tool. Two popular methods are deep learning algorithms (Soetasad & Fernando, 2025; Tan et al., 2021) and lexicon-based analyses (Bergsma et al., 2020; Yuan et al., 2025). Due to feasibility, this study will focus on the latter.

Bergsma et al. (2020) created a sentiment analysis on in-game character dialogue of the popular game The Elder Scrolls V: Skyrim (henceforth; Skyrim) based on Extended Affectionate Norms for English Words (henceforth; E-ANEW). To specify the analysis to Skyrim, the researchers compared the E-ANEW list to a game-specific word list, based on E-ANEW and an additional list from open access game material such as in-game books and character dialogue scripts. The game-specific lexicon performed slightly better on dominance and arousal aspects of discourse.

Panagiotopoulos et al. (2019) have done a sentiment analysis of Skyrim reviews. They justifiably note that game opinions are not a matter of positive or negative. Instead, opinions are based on *aspects* of the game, such as the graphics, audio and gameplay of the game (aspect-based approach). Using the Valence Aware Dictionary and sEntiment Reasoner (VADER; Hutto & Gilbert, 2014), the authors assign a sentiment score for different aspects of the game.

The discussed studies are mainly focused on methodology. Less attention has been paid to comparison of games within a genre or franchise. Sequels of a game allow controlled comparison, because they are of the same genre, developer and target audience. Differences in reviews between a game and its sequel may be attributed to changes in player perception and discourse over time. The present study therefore attempts to describe changes in sentiment and evaluative language in user reviews between *Kingdom Come: Deliverance* and its sequel, *Kingdom Come: Deliverance II* (henceforth KCD1 and KCD2, respectively).

KCD1 (2018) and KCD2 (2025) were created by Warhorse Studios. It is a one-player role player game, set in 15th century Bohemia. The franchise is considered unique in its kind, as it focuses on accuracy of medieval life. This study focuses on this game franchise, as its distinctive design may attract a relatively homogeneous audience, which supports a more controlled comparison of evaluative language across the two corpora.

The first objective of this study is to compare keywords in the reviews of each game. Building on the aspect-based approach of Panagiotopoulos et al. (2019) and the use of game-specific lexicon-based methods as in Bergsma et al. (2020), it analyzes differences in sentiment and the use of evaluative language across different game aspects. In addition, sentiment is examined at both the review and sentence levels in order to capture both overall evaluations and variations within reviews. Individual reviews may contain both positive and negative statements, and this may be better accounted for in a sentence-based analysis.

METHODOLOGY

Using the Python package *steamreviews (Wok, 2018/2026)*, English user reviews were collected from the Steam platform from *Kingdom Come: Deliverance I* (KCD1; 80,216 unique reviews) and *Kingdom Come: Deliverance II* (KCD2; 78,428 unique reviews) by Warhorse Studios (2018; 2025).

The raw data was preprocessed using *NLTK* for tokenization (Bird et al., 2009/2009), *spaCy* for lemmatization and part-of-speech (POS) tagging (Honnibal et al., 2014/2020) and *pandas* for data manipulation (McKinney, 2010). After removing empty or missing reviews, the final corpora consisted of 79,932 (KCD1) and 78,152 (KCD2) unique reviews. All text was converted to lowercase. URLs, punctuation and special characters were removed. Reviews were tokenized into words using NLTK and then lemmatized using spaCy. English stopwords were removed, however, 'not' was retained to preserve negation information. Finally, reviews were split into sentences for sentence-level sentiment analysis. Both versions of the tokens (with and without stopwords), the original, unprocessed review and the sentences were stored for different types of analysis.

All tokens, including stop words, were annotated with part-of-speech tags, using *spaCy.* Word frequency distributions were calculated for both corpora, excluding stopwords, using relative

frequencies per 1000 tokens. This allowed for more meaningful comparison between corpora by controlling for different sizes.

Adjectives, adverbs and verbs were extracted from the POS-tagged data to analyze evaluative language. To improve the identification of evaluative items, words were first lemmatized using WordNet, after which their sentiment polarity was determined using SentiWordNet 3.0 (Baccianella et al., 2010). For each word, an average sentiment score was calculated across its WordNet synsets, and only words with a polarity score above a predefined threshold ($|score| \geq 0.05$) were retained. This allowed for the exclusion of semantically neutral items. Frequencies were normalized per 1000 tokens.

Collocations were extracted for predefined aspect-related keywords, following the method of Panagiotopoulos et al. (2019). Several words were added to their list, to attempt a similar game-specific approach of Bergsma et al. (2020). The full list can be seen in Table 1. The collocations (window size = 2) were filtered by POS (adjective, adverbs, nouns and verbs) and the top five collocates per category were identified. Sentiment scores were then calculated for each aspect using the VADER sentiment analyzer. For each aspect, mean sentiment and standard deviation (SD) were calculated based on collocate frequencies.

Table 1. *Aspect categories and respective keywords, based on Panagiotopoulos et al. (2019).*

Aspect	Keywords
Gameplay	mission, item, map, weapon, mode, multiplayer, quest, combat, story, gameplay
Graphics	graphic, graphics, visual, look, aesthetic, animation, frame, design, art, world
Audio	audio, sound, music, soundtrack, melody, voice, valta
Community	community, support, toxic, friendly, player, players, developer, devs
Performance	server, bug, bugs, connection, lag, latency, ping, crash, crashes, glitch, glitches, fps, framerate

In addition to aspect-level sentiment analysis, VADER was used to perform sentiment analysis at both the review and sentence levels. Review-level sentiment provides a general assessment of a user's overall sentiment of the game. Reviews, however, may contain certain mixed opinions, where users praise certain aspects but may criticize others in different sentences. To capture this internal variation, another sentiment analysis was performed on sentence-level. To measure this variation, standard deviations were calculated for the review sentiment and sentence sentiment. It is expected that sentence-level sentiment means will be less positive than review level sentiment means, and that sentence-level sentiment standard deviations are higher than review level sentiment standard deviations, as a reflection of greater variation. Furthermore, the proportion of positive (score >0) and negative (score <0) reviews was calculated for each corpus. Intermediate results (e. g. tokenized data, POS tagging, frequency counts) were cached as pickle files for more efficient performance and reproducibility.

RESULTS

Table 2 presents the top 20 most frequent words in each corpus, showing that both KCD1 and KCD2 share many core terms such as *game*, *good*, *play*, and *story*, which indicates a similar focus on gameplay and overall experience. Table 3 focuses on the overlapping words from Table 2. The upper part of the table contains words that occur more frequently in KCD1, such as *not*, *get*, and *combat*, suggesting a stronger emphasis on critique. In contrast, the lower part contains words that are more frequent in KCD2, including *good*, *feel*, and *story*, which points to a more positive and experience-oriented evaluation.

Keywords unique to KCD1 include *really*, *bug*, *go*, and *system*. The presence of *bug* and *system* suggests a stronger focus on technical aspects and performance issues, while *really* may function as an intensifier, reinforcing evaluative statements. In contrast, keywords unique to KCD2 include *first*, *hour*, *love*, and *world*. These words point toward a more experiential focus, with *world* referring to the game environment and *love* indicating strong positive sentiment. Words such as *first* and *hour* may reflect discussions of early gameplay experience and time investment.

Table 2. *Top 20 most frequent words in KCD1 and KCD2 reviews (per 1000 tokens).*

	KCD 1		KCD 2	
#	**Word**	**Freq.**	**Word**	**Freq.**
1	game	45.67	game	48.95
2	not	26.54	not	23.20
3	get	11.61	good	12.53
4	like	10.45	play	11.86
5	good	10.44	first	10.38
6	play	10.18	like	10.25
7	combat	9.82	one	9.81
8	time	8.64	story	8.92
9	story	7.31	get	8.87
10	one	7.01	feel	8.14
11	make	6.82	time	7.69
12	really	5.88	well	7.66
13	great	5.84	make	7.54
14	well	5.83	great	7.22
15	quest	5.74	combat	6.71
16	bug	5.59	quest	5.39
17	feel	5.30	hour	5.30
18	go	5.03	love	5.28
19	even	4.79	even	5.05
20	system	4.65	world	4.98

Table 3. *Overlapping words from Table 2 with relative frequency per 1000 tokens, ordered by difference between KCD1 and KCD2.*

Word	KCD1	KCD2
not	**26.54**	23.20
get	**11.61**	8.87
combat	**9.82**	6.71
time	**8.64**	7.69
play	**10.18**	11.86
like	**10.45**	10.25
quest	**5.74**	5.39
game	45.67	**48.95**
good	10.44	**12.53**
feel	5.30	**8.14**
one	7.01	**9.81**
story	7.31	**8.92**
make	6.82	**7.54**
great	5.84	**7.22**
well	5.83	**7.66**
even	4.79	**5.05**

Tables 4 and 5 show the top 20 evaluative adjectives, adverbs and verbs in reviews of KCD1 and KCD2, respectively, per 1000 tokens. These data suggest that both corpora are dominated by positive evaluative language, with high-frequency adjectives such as *good*, *great*, and *amazing*, and verbs such as *recommend* and *enjoy*. In KCD2, strongly positive items such as *good*, *great*, and *love* occur more frequently, indicating positive opinions. In contrast, KCD1 contains relatively more items

that point to critical opinions, such as *bad*, *need*, and *fix*, suggesting that players more often frame their opinions in terms of issues or required improvements. KCD2 reviews show a stronger tendency toward positive opinions, whereas KCD1 reviews reflect slightly more critical evaluative language. Words like *think* and intensifiers such as *really* and *very* are common in both corpora, which shows that players use similar ways of expressing their opinions, even if the overall tone differs.

Table 4. *Top 20 evaluative adjectives, adverbs, and verbs in reviews of KCD1 per 1000 tokens, filtered using SentiWordNet.*

#	Adjective	Freq.	Adverb	Freq.	Verb	Freq.
1	good	9.76	just	8.52	want	3.54
2	great	5.80	very	6.32	love	2.85
3	other	3.15	really	5.88	think	2.52
4	bad	2.85	well	3.71	like	2.51
5	realistic	2.33	still	3.11	recommend	2.46
6	amazing	2.16	too	2.68	need	2.27
7	first	2.16	as	2.12	enjoy	2.25
8	main	2.13	much	1.85	buy	2.23
9	hard	1.71	actually	1.70	learn	2.11
10	worth	1.50	pretty	1.70	know	1.93
11	little	1.43	never	1.58	use	1.63
12	interesting	1.31	all	1.02	wait	1.59
13	easy	1.27	away	0.95	seem	1.33
14	beautiful	1.26	absolutely	0.93	spend	1.04
15	open	1.16	definitely	0.90	fix	1.04
16	big	1.14	often	0.75	lose	1.00
17	difficult	1.12	enough	0.74	die	0.96
18	long	1.12	truly	0.72	hit	0.90
19	nice	1.10	highly	0.72	pick	0.84
20	different	1.07	right	0.71	mean	0.75

Table 5. *Top 20 evaluative adjectives, adverbs, and verbs in reviews of KCD2 per 1000 tokens, filtered using SentiWordNet.*

#	Adjective	Freq.	Adverb	Freq.	Verb	Freq.
1	good	11.59	just	9.31	love	4.03
2	first	9.55	very	5.08	want	3.13
3	great	7.18	well	5.01	think	2.41
4	amazing	3.56	really	4.45	enjoy	2.26
5	other	2.55	still	3.42	recommend	2.24
6	main	2.35	too	2.60	like	2.22
7	bad	2.08	much	2.42	need	1.75
8	easy	1.54	as	2.08	know	1.73
9	worth	1.51	never	1.70	buy	1.68
10	hungry	1.43	absolutely	1.56	improve	1.43
11	beautiful	1.40	actually	1.46	wait	1.27
12	big	1.30	truly	1.21	spend	1.14
13	long	1.25	pretty	1.10	use	1.07
14	open	1.23	already	1.07	learn	1.07
15	little	1.17	definitely	0.92	seem	0.94
16	fantastic	1.14	highly	0.92	lose	0.91
17	hard	1.07	all	0.91	hit	0.77
18	realistic	1.04	easily	0.78	happen	0.67
19	perfect	0.99	yet	0.78	wish	0.65
20	interesting	0.94	away	0.77	pick	0.60

To gain more insight into how specific aspects of the game are evaluated, a collocation analysis was performed. For each predefined aspect (see Table 1), the most frequent collocates within a window of two words were extracted and analyzed by part-of-speech category. These collocates were then analyzed using VADER, and resulted in Table 6. VADER assigns a score between -1 (negative sentiment) and 1 (positive sentiment).

Table 6: *Mean (SD) sentiment scores per aspect for reviews of KCD1 and KCD2.*

Aspect	KCD1	KCD2
Gameplay	0.025 (0.200)	0.041 (0.201)
Graphics	0.034 (0.197)	0.050 (0.199)
Audio	0.034 (0.197)	0.051 (0.195)
Community	0.021 (0.183)	0.035 (0.185)
Performance	0.021 (0.184)	0.036 (0.188)

Table 6 shows that all aspects receive overall positive sentiment scores in both corpora. The values appear relatively neutral because they are approaching 0. However, most words in the corpus are neutral, so a VADER score will lean toward 0, and must therefore not be interpreted as a linear scale. Across all aspects, KCD2 consistently shows higher mean sentiment scores than KCD1, suggesting a more positive evaluation of the game in the newer release. This pattern is especially visible for graphics and audio, which receive the highest sentiment scores in both corpora and show the largest increase in KCD2. This may indicate that improvements in these areas are particularly noticeable to players.

In contrast, gameplay, community, and performance show lower sentiment scores overall, and although they also increase in KCD2, the differences are smaller. Notably, performance and community have the lowest scores in both corpora. The relatively similar standard deviations across all aspects and both corpora indicate that the variation in sentiment is comparable, meaning that player opinions are similarly spread out despite the overall shift towards more positive evaluations in KCD2.

Overall, the results suggest a general improvement in how different aspects of the game are perceived, with the largest improvements in technical and audiovisual features, while some areas, particularly performance and community, continue to receive more mixed evaluations.

Table 7 compares sentiment at the review and sentence levels in order to capture both overall evaluations and variation within reviews. While review-level sentiment reflects a player's general judgement, sentence-level sentiment allows for the identification of mixed opinions within individual reviews. Recall that it was expected that sentence-level sentiment would be less strongly positive than review-level sentiment, as players often combine praise and criticism within the same review. The results support this expectation. At the review level, both corpora show relatively high mean sentiment scores and a large proportion of positive reviews, particularly for KCD2 (M = 0.424, 86.5%) compared to KCD1 (M = 0.361, 78.5%). At the sentence level, both the mean sentiment and the proportion of

positive cases are slightly smaller than the sentiment scores on the full review (KCD1: M = 0.142, 65.7%; KCD2: M = 0.223, 74.8%).

The relatively high standard deviations in all conditions further indicate that sentiment varies considerably across reviews and sentences. Unexpectably, sentiment scores for reviews of both KCD1 and KCD2 had a higher standard deviation, and therefore more variation, than sentiment scores on individual sentences. This is further discussed in the next section.

Table 7. *Mean sentiment scores (SD) and proportion of positive reviews and sentences for KCD1 and KCD2.*

		KCD1	**KCD2**
Review level	Mean (SD) sentiment	0.361 (0.571)	0.424 (0.489)
	% positive	78,5%	86,5%
Sentence level	Mean (SD) sentiment	0.142 (0.462)	0.223 (0.438)
	% positive	65,7%	74,8%

DISCUSSION

Overall, the results indicate that KCD2 is evaluated more positively than KCD2 on the Steam platform. This is true for all studied levels of analysis, including keyword frequency, evaluative language, aspect-based sentiment, and both review- and sentence-level sentiment. This general conclusion may be attributed to a better player experience of all aspects of the sequel game (gameplay, audio, graphics, community and performance).

Importantly, the different levels of analysis show very similar patterns and are consistent in findings. The keyword analysis, evaluative language and aspect sentiment point in the same direction as the sentence-level and review-level insights.

It is evident, however, that the player evaluations are not uniform. Although it was expected that sentence-level sentiment would show greater variation than review-level sentiment, this is not clearly reflected in the results. One explanation is that many individual sentences are neutral or only mildly evaluative, which pulls sentiment scores towards zero and reduces overall variation. In contrast, review-level sentiment reflects an overall judgement, which is often more strongly positive or negative and therefore more extreme. In addition, splitting reviews into sentences may 'dilute' stronger opinions, as evaluative statements are distributed across multiple smaller units. As a result, sentence-level sentiment appears less variable than initially expected.

Although the results of the different analyses show great consistency, a number of limitations need to be noted regarding the present study. The first limitation is the efficiency of the used lexicons and Python modules in this study- VADER, spaCy and SentiWordNet are based on predefined word lists, which may not fully capture the context-dependent meaning of words in game reviews. Certain expressions, such as sarcasm, may be misinterpreted. SentiWordNet was used to identify evaluative

language, however, as can be seen in tables 5 and 6, several reported words are not evaluative, such as *other, main*, and *first*.

Although some game-specific words were added for the collocation-based aspect analysis (like *Valta*, the name of the game music composer), the list is kept small and comprehensible on purpose for the feasibilty of this study. Some important keywords used for the collocation analysis may have been missed. Furthermore, it is not guaranteed that the included collocations provide informative results. An in-depth investigation on the most effective keywords was considered beyond the scope of this study.

Another limitation is the arbitrary window size of 2 for the collocation analysis. Different window sizes were tested in the initial analyses. Judged subjectively, a window size of 1 gave little semantically informative collocates, and a window size of 3 included noise. Future analyses could further investigate ideal window sizing for collocates.

Future studies could use deep learning methods for sentiment analysis rather than lexicon-based methods. Additionally, it would be interesting to see how sentiment relates to the number of hours spent playing the game at the time of writing the review. This can easily be retrieved from Steam, along with other player data. In addition, future studies could include multiple game franchises to determine whether the observed patterns are consistent across other game genres. Finally, it may be interesting to see how sentiment evolves over time by comparing reviews from different stages after the release of a game.

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